# B1 fusion with 300-word output cap: Methodology and Evidence

CS781 Capstone – Krishna K Karthikeya G N

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**Status.** Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

## Research question

Inspect whether the same B1 fusion model loses overlap by generating summaries longer than the reference style. The cap was chosen on 32 held-out validation cases using an approximate ROUGE-1 F1 proxy, then checked on 64 confirmation cases.

## Architecture and mechanism

The existing B1 fusion generations are truncated to at most 300 whitespace-delimited words. No new model weights are trained for this entry.

## Measured result

| | |
|---|---|
| Scored submission | 1030821 |
| Codabench BERTScore F1 | 0.7783 |
| Archived local metric | not reported |

The proxy rose from 0.22108 to 0.25564 on validation and from 0.23031 to 0.26328 on confirmation; Codabench BERTScore F1 was 0.7783 versus 0.7674 for the untruncated export.

## Limitations and next check

The selection proxy was not BERTScore, and this is a decoding/export variant, not another architecture. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

## Source and reproducibility

Original source modules: b1_fusion.py, train_phase6.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

## Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.